

Technology Transfer of Open-Source Data Infrastructure for Embodied AI The AIRSPEED Case

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Abstract

Embodied AI depends on high-quality physical interaction data, yet such data remains expensive, fragmented, and difficult to reuse across robot platforms. This article presents AIRSPEED as a technology-transfer case study of how an open-source data production platform for embodied AI was developed within the Shenzhen Institute of Artificial Intelligence and Robotics for Society and advanced toward industrial deployment. AIRSPEED integrates teleoperation interfaces, heterogeneous robot adapters, simulation connectors, data collection, data generation, and dataset construction into a modular infrastructure for embodied AI data production. The case shows that technology transfer in emerging AI fields is not merely the transfer of intellectual property, but the coordinated transfer of architecture, engineering know-how, ecosystem relationships, customer validation, and commercialization logic. More broadly, the article argues that research institutions can improve deep-tech commercialization by starting from shared infrastructure bottlenecks, building reusable platform assets, using open source to accelerate adoption, and aligning early pilots with scalable business models.

1. Introduction: Why Embodied AI Needs Data Infrastructure

Embodied AI changes the data problem in artificial intelligence. Text, images, and videos can be collected from the digital world at large scale, but robot interaction data must be produced through physical execution [1], [2]. A useful dataset may include camera streams, robot trajectories, force and tactile signals, teleoperation commands, task outcomes, and environmental context. These records are costly because they depend on hardware, operators, calibration, safety checks, and repeated trials. As a result, many embodied AI systems still rely on small demonstrations, isolated laboratory datasets, or task-specific data pipelines.

AIRSPEED was motivated by a recurring engineering problem in embodied AI data production [3], [4]. When a team changed from one robot arm to another, or from a mobile robot to a humanoid platform, much of the data pipeline had to be rebuilt. Sensor streams, control commands, coordinate frames, timestamps, task labels, and teleoperation records were often stored and processed differently. This made data collection slow, but the larger problem came after collection: teams still had to clean, align, label, convert, version, and inspect the data before it could be used for training

or evaluation. AIRSPEED was created to make this process repeatable across robot embodiments.

The project therefore focused on the layer between robot hardware and model training. Instead of building another robot or another task-specific dataset, AIRSPEED connects teleoperation devices, robot adapters, simulation tools, and dataset construction workflows. The goal is practical: once a team has collected real interaction data, the same platform should help turn it into training-ready datasets, expand it through controlled simulation, and reuse it across related tasks or robot platforms. This made AIRSPEED suitable for technology transfer because the value was not limited to one laboratory prototype; it lay in a workflow that other teams could adopt.

Open source played a specific role in this process. It allowed outside researchers and developers to inspect the interfaces, test the data format, and build adapters for their own devices or robots [5]. This matters in embodied AI because no single institution controls all robot embodiments, sensors, simulators, and teleoperation tools. A closed platform may work within one vendor's system, but it is less likely to become a shared workflow. AIRSPEED's open-source strategy was therefore not only a dissemination choice; it was a way to test whether a common data-production interface could attract users beyond the original development team.

This article examines AIRSPEED as a technology-transfer case. It asks how a research institute can move from a technical prototype to an industrially usable infrastructure project when the core asset is not a single algorithm, but a reusable engineering workflow [3], [4]. The following sections discuss the problem AIRSPEED addressed, the platform architecture, the role of AIRS in incubation, the transition from technical validation to venture formation, and the lessons that may apply to other deep-tech transfer projects.

2. AIRS as an Incubation Platform for Technology Transfer

AIRSPEED began inside AIRS because the problem was still too early and too engineering-heavy to be solved as a conventional software startup. The team first had to understand how different robots, teleoperation devices, simulators, and data formats could be connected in one workflow. This required access to hardware, engineers, industrial partners, and testing scenarios. AIRS provided that environment before the project was ready to become a market-facing venture. This role is consistent with the broader view that universities and research institutes increasingly support a “third mission” beyond research and education: translating research outputs into economic and social impact through technology transfer and commercialization [7].

AIRS is one of Shenzhen's major public research institutions, established to connect frontier research with industrial deployment in artificial intelligence and robotics. Affiliated with The Chinese University of Hong Kong, Shenzhen, AIRS operates as a new type of research institute that combines academic research, engineering development, industry collaboration, venture incubation, and technology transfer. This institutional model is particularly relevant for embodied AI, where successful commercialization depends not only on algorithms, but also on hardware access, engineering integration, industrial partners, pilot scenarios, and standardization pathways.

Within this environment, AIRSPEED was developed as an open-source data production platform for embodied AI. Its technical objective is to reduce the cost and fragmentation of embodied AI data production through a modular architecture that connects teleoperation devices, heterogeneous robots, simulation environments, and dataset construction pipelines. Rather than building another vertically integrated robot system, AIRSPEED focuses on the missing horizontal layer: the data infrastructure that allows different robots, sensors, operators, and simulation tools to contribute to reusable embodied AI datasets.

AIRS helped AIRSPEED in three concrete ways. First, it gave the team an engineering environment in which data alignment, simulation interfaces, and robot integration could be tested before the platform had to satisfy paying customers. Second, it exposed the team to industrial users who cared less about the elegance of the software architecture and more about whether the system could reduce repeated integration work. Third, it helped the team translate the prototype into a commercialization plan: which parts should remain open source, which services could become enterprise offerings, and what kind of investors or partners would understand an infrastructure-style project.

AIRSPEED followed an internal incubation pathway. Under this model, the technical team develops the core technology within the institute, validates the system through engineering and pilot projects, and then establishes an independent company once the technology and market opportunity become sufficiently clear. This model reduces the early risk of commercialization: the project can rely on institutional resources during the uncertain prototype stage, while later gaining the autonomy needed for venture financing, customer delivery, and product scaling.

This incubation pathway is important because embodied AI infrastructure is difficult to commercialize through a simple licensing model. Much of the value lies in tacit engineering knowledge, interface adaptation, customer feedback, dataset workflows, and ecosystem trust. AIRSPEED's transfer therefore required more than moving intellectual property from a laboratory to a company. It required moving a team, a technical architecture, partner relationships, and a market thesis together.

The AIRSPEED case suggests that infrastructure-oriented AI projects often need an institutional bridge before they can become companies. The early asset is rarely just source code. It also includes interface knowledge, hardware adaptation experience, user feedback, testing environments, and trust from partners. AIRSPEED's transfer therefore involved moving a working engineering practice, not simply transferring a piece of intellectual property.

3. The AIRSPEED Platform: Architecture for Scalable Data Production

AIRSPEED treats embodied AI data production as an engineering workflow rather than a one-time collection task [6]. A demonstration collected on one robot becomes more valuable if it can be synchronized, checked, converted, replayed, expanded in simulation, and reused in later training cycles. The platform was therefore designed around the steps that make data reusable: device

connection, robot adaptation, simulation linkage, data cleaning, dataset construction, and version control.

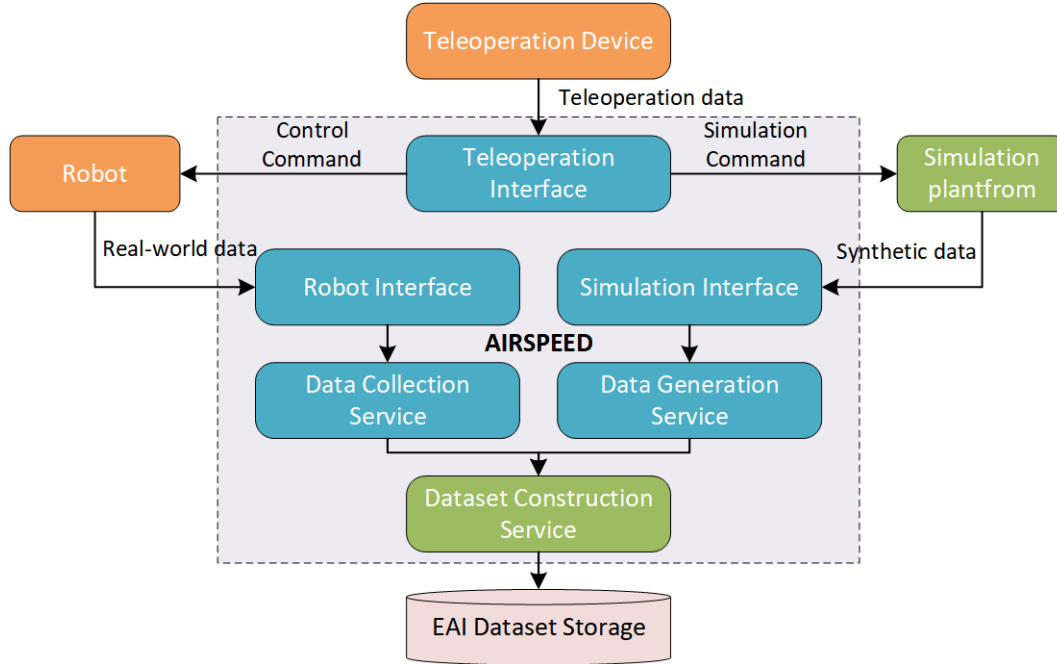


Figure 1. AIRSPEED architecture design

As illustrated in Figure 1, the platform is organized around three interfaces and three services. The three interfaces connect AIRSPEED to the key sources of embodied AI data: teleoperation devices, robot embodiments, and simulation environments. The teleoperation interface supports devices such as VR headsets, motion-capture systems, grippers, and other control terminals. The robot interface allows heterogeneous platforms, including robotic arms, mobile robots, quadrupeds, and humanoids, to be integrated through adapters. The simulation interface connects the platform to environments such as NVIDIA Isaac Sim and MuJoCo, allowing real-world demonstrations to be expanded through synthetic variation and simulation-based data generation.

As illustrated in Figure 2, on top of these interfaces, AIRSPEED provides three core services: data collection, data generation, and dataset construction. The data collection service receives real-world interaction data and performs synchronization, timestamp alignment, and initial cleaning. The data generation service uses simulation and data-derivation methods to expand limited real-world demonstrations into more diverse samples. The dataset construction service converts raw multimodal signals into structured datasets through cleaning, annotation, format conversion, quality control, and version management.

The key architectural contribution of AIRSPEED is the integration of these components into a closed-loop data production process. Real-world demonstrations serve as high-value seed data. These data are standardized and converted into training-ready datasets. The datasets are then used for model training, simulation rollout, and further data generation. Newly generated data are merged with real-world data, creating larger and more diverse datasets for the next training cycle. This

mechanism forms a data flywheel in which each round improves dataset scale, diversity, and reusability while reducing the marginal cost of additional data production.

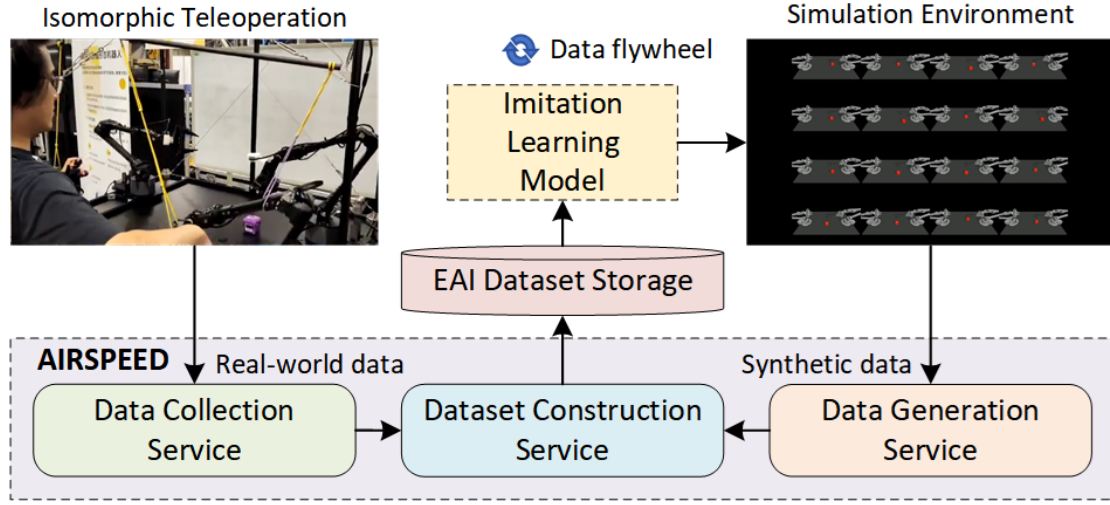


Figure 2. Services provided by AIRSPEED

4. AIRSPEED’s Incubation-Stage Development

During incubation, AIRSPEED’s main task was to turn a research idea into a repeatable engineering capability. The team did not begin with a finished product. It began with a practical problem: embodied AI data pipelines were repeatedly rebuilt whenever the robot, teleoperation device, simulator, or task changed. The incubation stage therefore focused on accumulating reusable technical assets rather than delivering one-off demonstrations. These assets included device interfaces, robot adapters, simulation connectors, data-alignment methods, dataset construction workflows, and early reference examples.

The first capability of AIRSPEED was hardware and device compatibility. AIRSPEED had to connect different sources of embodied AI data, including teleoperation devices, robot embodiments, and simulation environments. This required work that is often invisible in a research prototype: driver integration, control-signal conversion, timestamp alignment, coordinate-frame mapping, sensor synchronization, and interface testing. The value of this work was cumulative. Each new adapter made the next integration easier and helped turn scattered engineering fixes into a reusable compatibility layer.

The second capability was a modular data-production pipeline. AIRSPEED separated teleoperation interfaces, robot adapters, simulation connectors, data collection, data generation, and dataset construction into different parts of the platform. This separation mattered because customer scenarios rarely use the same hardware stack. A modular architecture allowed the team to change one part of the workflow without rebuilding the whole system.

The third capability was simulation-assisted data expansion. Real robot demonstrations remained the anchor, but they could be extended through simulation and data-derivation methods.

This was not intended to replace real data. Its purpose was to increase variation, test related scenarios, and reduce the marginal cost of producing additional training samples.

The fourth capability was dataset construction. AIRSPEED treated raw robot logs as unfinished material. Before such data can support model training, it must be cleaned, synchronized, labeled, formatted, checked, versioned, and made searchable. By building workflows around these steps, AIRSPEED moved beyond data capture and became a data-production platform.

The fifth capability was interface discipline. Because AIRSPEED was designed as an open-source platform, the team had to make interfaces understandable beyond the original developers. This required cleaner SDK boundaries, reference workflows, documentation, and dataset templates. In this sense, open source was not only a release strategy; it forced the project to make its engineering assumptions explicit.

By the end of incubation, AIRSPEED was no longer only a software prototype. The team had produced concrete components that could be used again: robot adapters, teleoperation interfaces, simulation connectors, timestamp-alignment tools, data-cleaning scripts, dataset templates, and reference workflows. This changed the nature of the project. Its value was no longer tied to one demonstration, one robot, or one dataset. It came from the fact that the same components could shorten the next integration cycle. The key output of incubation was therefore practical: AIRSPEED gave the team a way to connect a new device, collect interaction data, clean and synchronize the records, and convert them into a dataset without rebuilding the whole pipeline. Whether this repeatable workflow could support customers and commercialization is the question addressed in the next section.

5. From Engineering Capability to Commercial Validation

The engineering work described in Section 4 did not automatically create a business case. AIRSPEED still had to show that these components solved problems customers faced repeatedly. The key question was not only whether the platform could connect robots, devices, simulators, and datasets, but whether it could reduce the time spent on repeated integration, data cleaning, synchronization, and format conversion. If a pilot project only produced a custom script, it remained a service project. If it produced a robot adapter, dataset template, deployment script, or quality-control workflow that could be reused in the next project, it strengthened the platform. In platform ecosystems, such reusable complements can increase the value of the core platform because external users and partners help extend its applicability across use cases [11].

The first question was whether the problem would appear again. AIRSPEED had to show that data production was not a one-time integration task. Each new robot, sensor setup, teleoperation device, simulator, or task could require new alignment, cleaning, conversion, and dataset-building work. This made the platform useful only if it could reduce the work needed for the next project, not just finish the current one.

The second question was what each pilot left behind. A pilot was valuable only when it

produced something that could be reused: a robot adapter, a teleoperation interface, a dataset template, a quality-checking script, or a deployment workflow. If the output was only a customer-specific script, the project moved AIRSPEED toward services. If the output became part of the common platform, the project made AIRSPEED stronger.

The third question was what should remain open and what customers would pay for. The open-source layer allowed users to inspect the code, test the interfaces, and decide whether AIRSPEED could fit their own robots and data workflows [9]. The commercial layer should focus on work that customers are less likely to handle alone: private deployment, integration with their own hardware, large-scale simulation runs, dataset management, technical support, and workflow customization [10].

The fourth question was what evidence could convince customers and partners. A vision for embodied AI data infrastructure was not enough. AIRSPEED needed working integrations, tested data workflows, pilot feedback, and examples showing that the same adapter, template, or cleaning process could be used again. These were stronger signals than a research demo because they showed that AIRSPEED could reduce repeated engineering work.

The core lesson is that a research prototype becomes commercially credible when it helps users avoid doing the same engineering work again and again. Section 4 showed what AIRSPEED built during incubation. This section explained why those components mattered outside the laboratory: they shortened repeated integration work, turned pilots into reusable modules, gave developers an open interface to test, and created service opportunities around deployment, data management, and technical support. At that point, AIRSPEED was no longer only a prototype. It became a practical workflow that other teams could use.

6. Risk Mitigation: Keeping AIRSPEED Scalable and Defensible

AIRSPEED's main risks come from execution. The project can fail not because the idea is wrong, but because it becomes too dependent on simulation, too customized for early customers, or too difficult for outside developers to adopt.

The first risk is over-reliance on simulation. AIRSPEED should treat real-world demonstrations as the calibration anchor and simulation as a controlled expansion tool. A simulation-generated dataset should be linked to real demonstrations, checked against real-robot execution when possible, and assigned a quality score before it is used for training. For high-precision manipulation, real data should remain the benchmark.

The second risk is becoming a customization service rather than a platform. Early customers will naturally ask for special adapters, special workflows, and special data formats. AIRSPEED should accept such work only when it strengthens the common platform. Each customer project should produce at least one reusable asset: a robot adapter, a device interface, a dataset template, a quality-control workflow, or a deployment script.

The third risk is weak adoption. Open source can reduce this risk, but only if the platform is easy to inspect, test, and extend. AIRSPEED should therefore prioritize reference examples, clean SDK interfaces, documentation, and small reproducible workflows. Official standards may take time, so the practical objective is to let developers use AIRSPEED-compatible formats before formal standardization is complete.

The fourth risk is customer trust. Embodied AI data can include sensitive videos, task records, operator behavior, and proprietary robot information. AIRSPEED should therefore support private-cloud or on-premises deployment, role-based access control, audit logs, encrypted storage, and clear data-governance rules. The business model should not depend on taking ownership of customer data.

In short, AIRSPEED should keep real data as the source of truth, use simulation to scale carefully, use open source to spread interfaces, use standards to build legitimacy, and use paid pilots to test whether the platform solves repeatable customer problems.

7. Summary: Success Factors and Forward Outlook

The AIRSPEED case shows that technology transfer in embodied AI is not simply a handoff from laboratory to company. The project had to show that the same data workflow could work across different robots, teleoperation devices, simulators, and task settings. AIRS played an important role by giving the team time to test adapters, debug data alignment, work with industrial users, and decide which parts of the system should be open source and which parts could support paid deployment.

The broader lesson is that deep-tech transfer should begin with a problem that users face more than once. AIRSPEED was valuable because teams repeatedly faced the same practical work: connecting new hardware, synchronizing sensor streams, converting control records, cleaning robot logs, and preparing datasets for training. Other technology-transfer projects can ask a similar question: does the research output help many users avoid rebuilding the same toolchain?

Looking forward, AIRSPEED’s challenge is to prove that it can support more projects without becoming a custom-service company [8]. This will depend on practical progress: adding more robot adapters, improving multimodal synchronization, making dataset construction easier to inspect, supporting private deployment, and turning repeated customer requests into standard software modules. If AIRSPEED can do this, its importance will go beyond one platform. It will offer a concrete example of how a research institute can turn open-source AI infrastructure into an industrial tool while keeping the system open enough for developers to adopt.

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Biography

Tiantian Zhang is currently a Senior Supervisor at the Embodied AI Center, Shenzhen Institute of Artificial Intelligence and Robotics. Her research has long focused on the practical integration of embodied AI algorithms with data engineering. She led the technology transfer of the AIRSPEED Embodied AI Data Production Platform and holds relevant patents in the fields of robotics and artificial intelligence.

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